Data Science And Machine Learning

ASSIGNMENT-2

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Submitted To,

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(1) How do you prune a decision tree?

→ A decision tree can continue to grow indefinitely choosing splitting features and dividing into smaller & smaller partitions until each example is perfectly classified or the algorithm runs out of features to split on.

→ However, if the tree grows overly large, many of the decisions it makes will be overly specific and the model will have been overfitted to the training data.

→ The process of pruning a decision tree involves reducing its size such that it generalizes better to unseen data.

→ one solution to this problem is to stop the tree from growing once it reaches a certain number of decisions or if the decision nodes contain only a small number of examples. This is called ~~exact~~ early stopping or pre-pruning decision tree.

→ An alternative called post-pruning involves growing a tree is too large, then using pruning criteria based on the error rates at the nodes to reduce the size of the tree more appropriate level.

→ This is often a more effective approach than pre-pruning because it is quite difficult to determine the optimal of a decision tree without growing it first.

(2) Explain Classification Rule with examples.

→ Classification rules represent knowledge in

example

→ they are specified in terms

that form a hypothesis stating that

that happens

→ the antecedent comprises certain combinations

value while the consequent specifies the class value to

if the rules conditions are met

Example → Antecedent

if aerial = "Yes" then

return class = "bird" → consequent

else

return class = "mammal"

end if

→ Classification rule learning algorithms utilize a he

known as separate and conquer

→ The process involves identifying a rule that covers

of examples in the training data, and then sep

this partition from the remaining data

→ As rules are added, additional subsets of data

separated until the entire dataset has been covere

more examples remain.

One Rule Algorithm (1R Algorithm)

→

values of the feature.

→ Then for each segment, the algorithm predict the majority class.

→ The error rate for the rule based on each feature is calculated, and the rule with the fewest errors is choosen as the one rule.